

Unsupervised Deep Spectral Basis Learning for Generalized Eigendecomposition and Spectral Embedding

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Abstract—Spectral embedding has been widely used in statistical learning and geometric processing. Existing deep neural networks (DNNs) construct nonlinear mappings from node descriptors to embedding coordinates, relieving the scalability and generalization problems in traditional eigendecomposition-based graph embedding. However, additional QR-based orthogonalization or affine transformation is required for training deep spectral embedding models. In this work, we introduce an unsupervised spectral basis learning (SBL) framework for generalized eigendecomposition of graph matrices, benefiting from linear graph convolutions (LGCs) for spectral embedding. We design a novel spectral embedding criterion to learn the spectral basis in an analogous way to iterative power deflation, avoiding additional QR-based orthogonalization or transformations for spectral bases estimation. However, the proposed approach enables the lining up of spectral bases between graphs, which relieves eigenvector switching and facilitates graph matching. We showcase the efficacy of the proposed SBL for generalized graph eigendecomposition with performance gains over state-of-the-art deep spectral embedding methods. The source code is available at <https://github.com/DYS2108/SBL>

Index Terms—Deep spectral basis learning (SBL), generalized eigendecomposition, graph matching, spectral embedding.

I. INTRODUCTION

GRAPH representation learning and graph spectral properties have been widely used in statistical analysis and geometric processing scenarios, such as clustering [1], [2], segmentation [3], and correspondence [4], [5]. Spectral methods require eigendecomposition of graph matrices, such as the graph Laplacian, as a combinational variant of the Laplace-Beltrami operation, with the worst computational complexity of $O(n^3)$ and the scalability problem regarding large-scale

graphs. The Nyström approximation algorithm employs matrix blocking to reduce the complexity from $O(n^3)$ to $O(pn \log n + p^3 + p^2n)$, where the block size $p \leq n$ in practice. Nevertheless, the prior sampled subblocks are required in advance to implement eigendecomposition, where sampling landmarks are limited to handling a sparse graph Laplacian when they are not in the neighborhood. Landmark sampling and subspace projection have been used for fast eigendecomposition of the graph Laplacian in functional map-based correspondence [6]. However, the per-graph sampling and subspace generation cannot be generalized to novel graphs.

Deep neural networks (DNNs) establish the mapping between node descriptors and the embedding coordinates for graph embedding [7], [8], [9], [10], [11]. The DNN handles the out-of-sample extension [12] by learning the spectral embedding network for the independent and identically distributed (i.i.d.) manifold data. The graph convolutional network (GCN)-based embedding used the Chebyshev basis functions to avoid the eigendecomposition for spectral graph convolutions [7], [8]. Especially, the simplified GCN removed the nonlinear activation function and the learnable weight (LW) matrix [9], which can be viewed as a variant of the iterative power method for the dominant eigenvector. However, the dominant eigenvector of the graph Laplacian is related to a constant eigenvalue and fails in spectral embedding [10]. Marin et al. [11] approximated the basis functions by using the PointNet and avoided the eigendecomposition in correspondence computation. The approximated basis computation is efficient in the online testing process, though there is a lack of orthogonality in the approximated basis functions and connections with the spectral embedding.

The generalized DNN-based eigendecomposition utilized paired eigenvalues and Rayleigh quotients for training [13]. The SpectralNet realized the eigendecomposition by using an alternate fully connected neural network (FCN)-based embedding and QR orthogonalization with a ratio-cut loss, which was optimized from large batches [14]. The SpecNet2 avoided orthogonalization in the training process and approximated the eigenvectors by an unconstrained optimization, where the batch data were required to solve an affine transformation in the online inference process [15]. In order to remove the inconsistency of the embedding coordinates in batch learning, the BASIS aligned embedding with the anchors in the

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training process [16]. The aforementioned methods required additional QR-based orthogonalization or affine transformation for eigendecomposition in the context of spectral clustering, where the embedding dimensionality equals the cluster numbers. Considering tasks in need of more eigenvectors, such as spectral matching with nodewise correspondence, more computational complexities are required, considering the QR-based orthogonalization and transformations.

In this article, we propose an unsupervised spectral basis learning (SBL) for generalized and computationally efficient eigendecomposition of the graph Laplacian. Specifically, we present a spectral embedding criterion to encourage graph embedding consistent with the eigenstructures of the graph Laplacian. We introduce a linear GCN-based spectral basis approximation module to be learned in an unsupervised manner. Unlike QR-based eigendecomposition of graph matrices, which is sensitive to shape scaling or bending, the proposed SBL is promising to achieve lining up eigenvectors between graphs, avoiding eigenvector switching and sign flipping problems using conventional methods. We provide a theoretical bound of the approximated spectral bases analogous to the power deflation. The proposed SBL enables a generalized eigendecomposition of the graph Laplacian. The approximated spectral bases facilitate correspondence computation in the embedding space. We have evaluated the proposed method on public datasets of FAUST [17], SCAPE [18], TOSCA [19], RibFrac [20], and manifold toy data [21] for the generalized eigendecomposition, with performance gains over state-of-the-art deep spectral embedding methods. The main contributions of this work are summarized as follows.

- 1) We present a linear graph convolution (LGC)-based network to uncover the eigenstructures of the graph matrices in an analogous way to the iterative power deflation, with a theoretical bound of approximated spectral bases.
- 2) The proposed SBL shows potential to uncover consistent spectral bases between graphs, enabling graph correspondence and relieving the eigenvector switching and sign flipping problems.
- 3) We have evaluated the proposed SBL on generalized eigendecomposition of graph matrices from public datasets. The experimental results demonstrate superior performance over the state-of-the-art deep spectral embedding methods.

II. RELATED WORKS

A. Fast Eigendecomposition

Landmark sampling and subspace projection have been used for fast eigendecomposition. Uniform sampling and k -means clustering-based landmark sampling were used for fast spectral clustering [22]. Variants of the Nyström algorithms have generalized the spectral embedding [23], [24]. Nyström extension methods and their applications in spectral embedding conducted a fast eigendecomposition on a small dataset, where the eigenvectors were extended to the whole dataset [25]. However, landmark sampling restricted the application of Nyström algorithm to sparse graph matrices,

considering the degenerated block matrix with nodes not in the neighborhood.

Random subspace projection has been used to approximate leading eigenvectors via power iteration and a singular value decomposition [26]. The subspace iteration alternated between inverse power iteration and orthogonalization for eigenvectors [27]. A random projection tree used precomputed centroids for fast spectral clustering [28]. Poisson-disk sampling has been used for fast eigendecomposition of the Laplace–Beltrami operator [6], where the eigendecomposition was restricted to the subspace. However, the graph-specific point sampling and subspace projection restricted the generalization capacity. Unlike the above per-graph sampling and subspace construction, we conduct the generalized eigendecomposition of graph matrices within the proposed deep SBL framework.

B. Deep Learning-Based Graph Embedding

The DNN enables computationally efficient online graph embedding by parameterizing a nonlinear graph embedding function [29], [30]. The graph embedding can be learned in an unsupervised manner to address distributions in the latent embedding space via a variational graph autoencoder [31] and adversarial training [32]. The graph embedding with the attention scheme exploited the graph topology to enhance representation learning [33], [34], [35], [36]. The spectra-free GCN utilized the first-order Chebyshev polynomial approximation in graph encoding [7]. The unsupervised GCN utilized the self-reconstruction objective regarding similarity matrices [31], [32] and feature matrices to learn the graph autoencoder [37], [38]. Priebe et al. [10] found that the unsupervised GCN was capable of estimating dominant eigenvectors of graph matrices. The classification task-specific supervised learning and QR iterations [39] can be used to estimate other eigenvectors, though they introduce enlarged computational complexities. The autoencoder-based diffusion network generated a mapping function between the data and the diffusion maps in a supervised manner [40]. The diffusion network required to compute the eigenvectors of the training data in the learning process, which posed difficulty in batch-based learning with large datasets. From a broad perspective, the spectral embedding falls in the representation learning for specific tasks. We propose a novel unsupervised SBL methodology, which demonstrates its superior generalization and scalability capacities in spectral embedding.

III. METHOD

We propose a linear GCN-based SBL framework to generate the underlying spectral embedding of the Laplacian matrix, as shown in Fig. 1. Given a graph $\mathcal{G}$, the goal is to build the spectral embedding space by inferring the spectral bases Z of the normalized graph Laplacian matrix $L = I - D^{-1/2}AD^{-1/2}$, where A denotes the affinity matrix, and D denotes the diagonal degree matrix. Given the node descriptor $F \in \mathbb{R}^{n \times q}$, the affinity matrix $A_{ij} = \exp(-\|F_i - F_j\|_2^2)/\nu$ and the normalized affinity matrix $\tilde{A} = D^{-1/2}AD^{-1/2}$. We utilize cascaded graph convolutions in an analog to the power iteration, where the refined node descriptor $\tilde{F} \in \mathbb{R}^{n \times \tilde{q}}$ serves as the initial vectors.

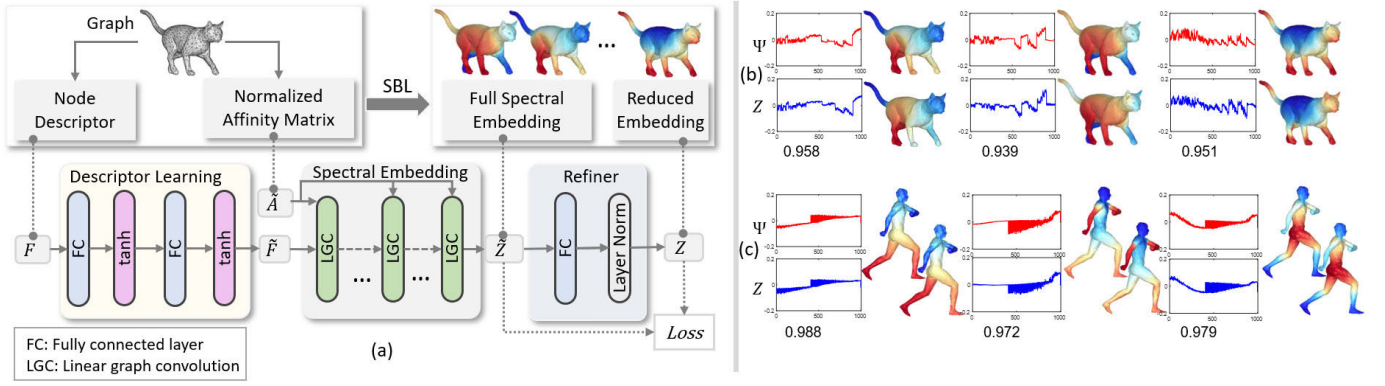


Fig. 1. (a) Architecture of the proposed SBL framework for generalized eigendecomposition and spectral embedding. The graph node descriptor F is fed to the FCN-based descriptor learning module. Then the descriptor $\tilde{F}$ combined with the normalized affinity matrix $\tilde{A}$ is used to predict the full spectral embedding $\tilde{Z}$ via cascaded LGCs in an analogous way to the power deflation. $\tilde{A}$ serves as the input of LGC layers. The SR further enforces the spectral embedding criteria for reduced spectral bases Z . Visualization of the predicted spectral bases Z and the ground truth Ψ in curves and pseudocolors on sampled meshes from (b) TOSCA and (c) FAUST datasets. The cosine similarity $z \cdot \psi$ is shown at the bottom.

The node descriptor learning is required to cover the spectral space for power iteration. Given the node embedding $\tilde{F}$ and normalized affinity matrix $\tilde{A}$, the spectral embedding module aims to discover the underlying spectral bases $\tilde{Z} \in \mathbb{R}^{n \times n}$ of A . The GCN-based spectral embedding function $\phi: (\tilde{A}, \tilde{F})_G \rightarrow \tilde{Z}$ maps the affinity matrix $\tilde{A}$ and the descriptor $\tilde{F}$ to spectral bases $\tilde{Z}$. To avoid possible correlations in approximated spectral bases $\tilde{Z}$, we introduce a spectral refiner (SR) to further enforce the spectral embedding criteria for reduced spectral bases Z .

In the online testing process, when given the normalized affinity matrix and graph node descriptor of a novel graph, the proposed SBL produces the spectral embedding. The resultant spectral bases are effective in relieving the eigenvector switching or sign flipping problems, facilitating dense graph correspondence.

A. Spectral Basis Learning

The proposed SBL benefits from the GCN to learn a generalized spectral basis approximation of multiple frequency bands, relieving the computational complexity in matrix eigendecomposition. When given the normalized affinity matrix and node descriptor, the vanilla GCN takes advantage of feature aggregation from the high-order neighboring nodes using cascaded graph convolutions. The nonlinear activation function has been used to speed up the convergence rate with contraction mapping on the linear operations of node embedding. Recent research has empirically demonstrated that the performance gains obtained by the GCN can be mostly ascribed to the feature aggregation [9], [41]. The LGC has shown computational efficiency and performance improvements by removing the nonlinear operations with the LGCs, and

$$Z^{(i)} = \tilde{A}^i Z^{(0)} \Pi_{i=1}^l W^{(i)} \quad (1)$$

where $Z^{(i)}$ denotes the embedding of the i th LGC layer. $W^{(i)} \in \mathbb{R}^{n \times n}$ is the LW matrix of the i th LGC layer. Considering the power iteration of the graph matrix $\tilde{A}$, the GCN is capable of finding the dominant eigenvector of the graph

matrix [10], [39]. Specifically, the linear GCN is the same as the power iteration to compute the dominant eigenvector by setting the weight matrix to an identity matrix. However, the unsupervised linear GCN can only recover the dominant eigenvector of the graph matrix and fails to find the spectral embedding with multiple frequency bands [39]. Additional supervision and QR iterations were required for spectral embedding. In this work, we present an LGC-based unsupervised SBL for end-to-end spectral embedding with multiple frequency bands.

1) *Training Loss*: We design a spectral embedding criterion to encourage the embedding consistent with the eigenstructures of the graph Laplacian matrix. Specifically, we impose orthogonality and fundamental properties of eigenvectors on Z : 1) $\tilde{A}z_i \parallel z_i, 1 \leq i \leq k$; 2) $\text{off}(Z^T \tilde{A}Z) = 0$; and 3) $z_i^T z_j = 0 (i \neq j)$. The spectral basis approximation loss $\mathcal{L}$ is defined as

$$\mathcal{L}(Z) = \mathcal{L}_{\text{ev}} + \alpha_1 \mathcal{L}_{\text{dia}} + \alpha_2 \mathcal{L}_{\text{ort}}. \quad (2)$$

We employ the eigendecomposition characteristics and design spectral embedding loss

$$\mathcal{L}_{\text{ev}} = \sum_i \left(\frac{(\tilde{A}z_i) \cdot z_i}{\|\tilde{A}z_i\|_2} - 1 \right)^2 \quad (3)$$

where z_i denotes the i th column vector of the approximated spectral basis matrix Z . Considering the definition of eigendecomposition, $AZ \approx Z\Lambda$, and the normalized vector Az_i is required to coincide with z_i . By minimizing the $\mathcal{L}_{\text{ev}}$, the approximated spectral bases satisfy the fundamental properties of eigenvectors. We also impose the diagonal similarity transformation constraint

$$\mathcal{L}_{\text{dia}} = \|\text{off}(Z^T \tilde{A}Z)\|_{\text{Fro}}^2 \quad (4)$$

where $\|\cdot\|_{\text{Fro}}$ denotes the Frobenius norm. $\text{off}(Z^T \tilde{A}Z)$ denotes the off-diagonal matrix with the elements of matrix $Z^T \tilde{A}Z$ that are not located on the main diagonal. Since matrix $\tilde{A}$ is symmetric, $Z^T \tilde{A}Z$ is required to be a diagonal matrix when Z approaches the eigenvectors of matrix $\tilde{A}$. The orthogonality constraint

$$\mathcal{L}_{\text{ort}} = \|Z^T Z - I\|_{\text{Fro}}^2 \quad (5)$$

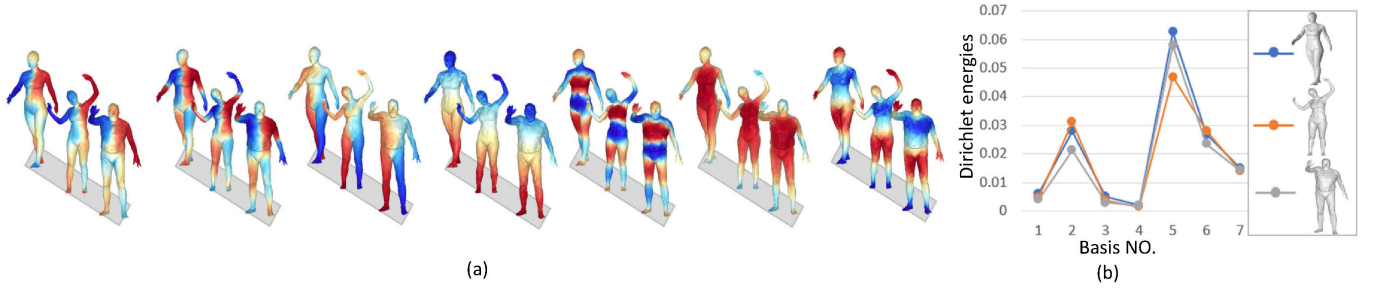


Fig. 2. (a) Seven approximated spectral bases visualized on three shapes in the FAUST dataset. Each subplot illustrates an approximated spectral basis visualized on three shapes. (b) Discrete Dirichlet energy $\gamma_i = z_i^T L z_i / |1 \leq i \leq k|$ associated with approximated spectral bases Z .

where I is the identity matrix. $\mathcal{L}_{\text{ort}}$ is used to ensure the orthogonality of the approximated spectral bases. The hyperparameters of α_1 and α_2 are used to balance the orthogonality and fundamental properties of eigenvectors.

Different from existing GCN-based spectral embedding [10], [39], our model finds spectral bases with multiple frequency bands via cascaded LGCs, relieving the computational complexity associated with the numerical QR iteration and additional supervision. The learnable matrix $W^{(i)}_{i=1,\dots,l}$ is optimized to deflate the node descriptor embedding by removing the dominant eigenvectors gradually. The proposed SBL is promising to find consistent embedding with a lined-up series of spectral bases, as shown in Fig. 2, being robust to eigenvector switching caused by shape scaling and bending [42]. However, the signs of the approximated spectral bases are consistent across graphs, avoiding random sign flipping in eigenvectors among graphs.

2) *Architecture*: Instead of using handcrafted node descriptors, we utilize two fully connected layers with tanh activation to parameterize the descriptor learning function $\theta : F_{n \times q} \rightarrow \tilde{F}_{n \times \tilde{q}}$. Since the spectral embedding module is built to perform a full eigendecomposition of the graph Laplacian matrix, we expect the input features to be information-rich and cover as many eigenvectors as possible. To achieve this, we adopt an expansion scheme to design the two fully connected layers with the weight matrix dimensionality of $q \times \tilde{q}$ and $\tilde{q} \times \tilde{q}$. In our experiments, $\tilde{q}$ is set to the node number n of the graph. The following LGC layers take the normalized affinity matrix $\tilde{A}$ and the descriptor $\tilde{F}$ as the input for spectral embedding. Considering that the refined node features $\tilde{F}$ are not guaranteed to cover the full set of eigenvectors, we impose the FCN-based SR for reduced spectral bases Z , and

$$Z = \tilde{A}^l \tilde{F} \Pi_{i=1}^l W^{(i)} \text{softmax}(W_r) \quad (6)$$

where $W_r \in \mathbb{R}^{n \times k}$ denotes the LW matrix of the refiner. The proposed SR can be viewed as a basis selector for k spectral bases from the full set of spectral bases. The proposed SR can be viewed as a basis selector for k spectral bases from the full set of spectral bases.

3) *Training Scheme*: Instead of imposing embedding criteria just to the output spectral bases, we introduce the spectral embedding constraint to both the full and the reduced spectral bases. First, we initialize the SBL by minimizing $\mathcal{L}(\tilde{Z})$ [see (2)] for optimized descriptor learning and spectral embedding

modules. Second, we optimize W_r of the refiner by minimizing $\mathcal{L}(Z)$, where parameters of the descriptor learning and spectral embedding modules are fixed. The training scheme is critical because optimizing only $\mathcal{L}(Z)$ would restrict the model's capacity to generate a limited set of eigenvectors. By prioritizing the full factorization in the first phase, we retain the flexibility to discover the underlying eigenstructure of the graph matrices and provide the specific number of approximated spectral bases for various tasks.

Note that the node descriptors are not guaranteed to lie in the space spanned by the full set of eigenvectors of the graph Laplacian matrix. In other words, $\tilde{F}$ can be orthogonal to a combination of a subset of eigenvectors. The column vectors of $n \times n$ -dimensional $\tilde{Z}$ tend to be correlated even under the orthogonal constraints. The conventional spectral embedding criterion, such as $((\tilde{A} \tilde{z}_i) \cdot \tilde{z}_i) / (\|\tilde{A} \tilde{z}_i\|_2 - 1)^2$, can be used to measure the confidence scores of the approximated spectral basis. However, it provides no guarantee for the orthogonality of the resultant approximated spectral bases. The additional QR decomposition has been used to compute the orthogonal spectral bases [39]. In contrast, the proposed SBL discovers spectral bases without additional scoring and QR iterations.

4) *Complexity*: The proposed SBL achieves computationally efficient eigendecomposition of the graph Laplacian matrix. Given the learned SBL with l LGC layers, the weight matrix multiplication $\Pi_{i=1}^l W^{(i)}$ and W_r can be precomputed. The spectral bases of a n -node graph are computed with a complexity of $O(n(lb^2 + \tilde{q}b + \tilde{q}k))$, where b and $\tilde{q}$ denote the bandwidth of affinity matrix $\tilde{A}$ and the channel number of the descriptor, respectively. The normalized graph Laplacian matrix tends to be a sparse matrix when we only consider the nearest neighbors in the affinity matrix with $b \ll n$. The proposed spectral embedding greatly relieves computational complexity in graph matrix eigendecomposition (see Fig. 3), which is promising to handle the scalability problem compared with the conventional eigendecomposition method.

B. Theoretical Analysis

The GCN involves power iteration regarding the graph matrix, which enables dominant eigenvector recovery from a normalized adjacency matrix [10] or a modular matrix [39]. Such behavior is viewed as the degeneration of the GCN, since the output embedding approaches a rank-1 matrix. The QR

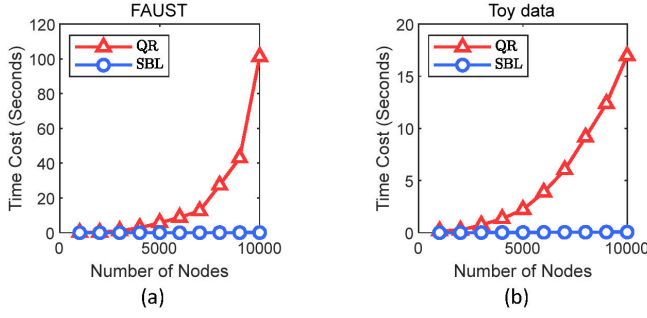


Fig. 3. Time cost of the proposed SBL and the conventional QR iteration method on graphs of different sizes on (a) FAUST and (b) toy data.

decomposition has been used to discover multiple eigenvectors [39]. In contrast, we present an LGC-based framework to discover spectral bases from the graph Laplacian matrix.

Let $\{\lambda_i | i = 1, \dots, k\}$ and $\{\psi_i | i = 1, \dots, k\}$ be eigenvalues and eigenvector of graph Laplacian matrix L , s.t. $L\psi_i = \lambda_i\psi_i$. The approximated spectral bases define the k -dimensional embedding coordinate $\{z_{1,i}, \dots, z_{k,i}\}^T | 0 \leq i \leq n$ to minimize $\text{tr}(Z^T LZ)$. The graph Laplacian matrix L bears the same eigenvectors as the normalized affinity matrix $\tilde{A}$. Unlike the GCN-based dominant eigenvector discovery from the normalized affinity matrix $\tilde{A}$, the proposed SBL draws inspiration from the power method with deflation and progressively removes dominant eigenvectors for eigendecomposition of the graph matrix. Considering the m th column vector z_m of the approximated spectral bases Z , the power iteration will converge to the m th eigenvector of $\tilde{A}$ when $z_m = \sum_{i=m}^n \beta_i \psi_i$. The LW matrix in the proposed SBL aims to remove the dominant eigenvectors progressively from matrix Z 's column vectors.

Proposition 3.1 (Annihilation Method [43]): Let $z = \sum_{j=i}^n \beta_j^{(i)} \psi_j$. If the power method is applied to z , $\lim_{l \rightarrow \infty} \tilde{A}^l z = \psi_i$. The proposed spectral basis approximation module produces the i th eigenvector of $\tilde{A}$.

The annihilation method (see Proposition 3.1) inspires the design of the SBL architecture. Our network employs multilayer LGCs to emulate the power iteration with deflation. The spectral embedding criterion is designed to enforce eigenvector properties and orthogonality, which ensures that the optimized weight matrix effectively approximates the behavior of iterative power deflation. The annihilation method requires the initial vectors to be nonzero. This condition is naturally satisfied since node features are used as the input. According to Proposition 3.1, the LW matrices $\{W^{(1)}, \dots, W^{(l)}\}$ in the SBL are optimized for progressive deflation, where column vectors of Z bear multifrequency spectral bases.

Proposition 3.2: Let δ_i be the ratio of the i th eigenvalue and the remaining eigenvalues of the normalized affinity matrix $\tilde{A}$, s.t. $\delta_i = (\max[|\lambda_{i+1}|, \dots, |\lambda_n|]) / \lambda_i$. ψ_i and z_i denote the i -th ground truth and approximated eigenvectors of $\tilde{A}$. If $z_i^T \tilde{A} z_i \geq \lambda_i \epsilon$ and z_i is orthogonal to $\psi_m | 1 \leq m < i$, $\epsilon > 0$, then

$$\langle \psi_i, z_i \rangle \geq \sqrt{\frac{\epsilon - \delta_i}{1 - \delta_i}}. \quad (7)$$

However, the iterative update of the approximated eigenvector by the power multiplication satisfies

$$\langle \psi_i, \tilde{A} z_i / \|\tilde{A} z_i\| \rangle \geq \langle \psi_i, z_i \rangle, \text{ and} \quad (8)$$

$$\langle \psi_i, \tilde{A} z_i / \|\tilde{A} z_i\| \rangle \geq \sqrt{\frac{\epsilon - \delta_i}{\epsilon - \delta_i - \epsilon \delta_i^2 + \delta_i^2}}. \quad (9)$$

Proposition 3.2 gives a bound [see (7)] for the approximated eigenvector estimated by the proposed SBL. The power iteration $\tilde{A} z_i$ improves the approximation accuracy of the spectral bases Z [see (8) and (9)], thereby providing theoretical support for the asymptotic convergence of SBL. The similarity between the spectral embeddings generated by SBL and the real ones increases progressively with the number of linear graph convolutional layers.

The proposed SBL exploits the LWs in the LGC for the power method with deflation. Different from per-graph eigendecomposition by conventional numerical methods, the proposed SBL provides an end-to-end generalized and multi-frequency spectral basis.

C. Implemental Details

We implement the proposed USL using the Tensorflow toolkit on a PC with an NVIDIA GTX 3090 GPU. The ADAM algorithm is used with a learning rate of $1e-3$. The momenta are set to 0.9 and 0.999. The mini-batch consists of two graphs with $n = 1000$ nodes. The proposed SBL contains $l = 5$ LGC layers for spectral basis estimation. The hyperparameters α_1 and α_2 in the loss $\mathcal{L}$ [see (2)] are set to 1 and 0.2, respectively. The training takes 35 epochs. The online spectral basis inference of a graph Laplacian matrix with 1000 nodes takes approximately 0.003 s.

IV. EXPERIMENTS

A. Datasets and Metrics

We validate the proposed SBL on manifold learning toy data [21], FAUST [17], SCAPE [18], and TOSCA [19], RibFrac [20], and Drive&Act [44] datasets.

The toy dataset includes corner planes, twin peaks, punctured spheres, 3-D clusters, and Gaussian randomly selected samples. There are 10k nodes for each type of toy data, and we generate 100 graphs by randomly sampling 1k nodes. The toy data are randomly split into the training data (90%) and testing data (10%). The input node descriptor includes the XYZ coordinate, the normal, and the fast point feature histograms with a channel number $q = 39$.

We validate the proposed SBL on 3-D mesh datasets of FAUST [17], SCAPE [18], and TOSCA [19]. We follow the data split protocol of FAUST [45] for model training and evaluations, where 20% of the training set is held out for validation. SCAPE [18] contains 71 meshes, which are only used for testing. We use 31 meshes of four kinds of animals, i.e., cats, dogs, wolves, and horses, in TOSCA [19], which are randomly divided into the training data (70%) and testing data (30%). We employ the quadric edge collapse decimation algorithm [46] for mesh re-sampling and simplification, where each graph contains 1k nodes. The input mesh node descriptor

has $q = 360$ channels, including the XYZ coordinate, the normal, the signature of the orientation histogram [47], the shape diameter function, and the curvature.

The RibFrac dataset [20] contains 660 chest CTs of patients with rib fractures, which are randomly split into 420 for training, 80 for validation, and 160 for testing. We generate a supervoxel graph of rib cages and lungs using SLIC [48]-based supervoxel decomposition and one-ring neighborhood. Supervoxel node features include a 50-bin HU histogram and a 3-D offset vector to the centroid. As to the rib cage graphs, we use a 4-D geodesic distance vector regarding four extreme boundary supervoxels on bounding planes with q of 57.

For the vehicle action recognition dataset Drive&Act [44], we use the same data splits and evaluation metrics as [49]. Each graph consists of 25 joints tracked across 90 consecutive frames, yielding a total of 2250 nodes per graph. Features include 3-D skeletal coordinates and temporal displacements across all frames, and the channel number $q = 183$.

We use average cosine distance (ACD) to evaluate the SBL. Given a pair of predicted and ground-truth spectral basis $\{(\psi_m, \psi_{m'})\}_{m=1}^k$

$$\text{ACD} = \frac{1}{k} \sum_{m=1}^k \psi_m \cdot \psi_{m'}. \quad (10)$$

The ground-truth spectral basis Ψ is obtained using QR iteration, and $\psi_{m'} = \arg \max_{1 \leq m' \leq n} \psi_m \cdot \psi_{m'}$. As to the graph correspondence between graphs $\mathcal{G}_r$ and $\mathcal{G}_t$, we measure the average geodesic distance (AGD). For a set of predicted and ground-truth matched point pairs $\{(s_{t,j}, s_{t,j}^*)\}_{j=1}^n$ associated with a reference point $s_{r,i} \in \mathcal{G}_r$ on the target graph $\mathcal{G}_t$

$$\text{AGD} = \frac{1}{n} \sum_{j=1}^n \frac{d_t(s_{t,j}, s_{t,j}^*)}{\sqrt{\text{area}(\mathcal{G}_t)}} \quad (11)$$

where $\text{area}(\mathcal{G}_t)$ denotes the area of graph $\mathcal{G}_t$, and d_t denotes the geodesic distance on shape $\mathcal{G}_t$. Generally speaking, high values of the ACD and low values of the AGD indicate reasonable spectral embedding and graph matching.

B. Spectral Basis Approximation

1) *Toy Data*: We evaluate the SBL on manifold embedding of well-known toy data, as shown in Table I. We validate the proposed method in two experimental settings, i.e., a *single* model for each type of toy data, and a *mixed* model for all five types of toy data. In both experimental settings, the proposed SBL is effective for calculating nontrivial spectral bases for manifold embedding. The ACDs on the testing data range from 0.93 to 1.00 on all toy data using the *single* setting. In the *mixed* setting, the proposed SBL learned from node samples from mixed toy data also shows generalization capacity regarding all types of toy data, with the ACD ranging from 0.83 to 0.99. The proposed SBL is feasible for discovering the intrinsic spectral embedding.

2) *Faust and Scape*: We conduct validation on FAUST and SCAPE datasets in two experimental settings: the models trained on FAUST and tested on FAUST (F-F) and

TABLE I
SPECTRAL BASIS APPROXIMATION ACCURACIES OF THE FIRST TWO NON-TRIVIAL SPECTRAL BASES IN TWO SETTINGS. SINGLE: ONE MODEL FOR EACH TYPE OF TOY DATA. MIXED: ONE MODEL FOR ALL FIVE TYPES OF TOY DATA. (CP: CORNER PLANES, PS: PUNCTURED SPHERE, TP: TWIN PEAKS, 3DC: 3-D CLUSTERS, GS: GAUSSIAN RANDOMLY SAMPLES)

		<i>single</i>		<i>mixed</i>	
		$\cos(z_1, \psi_{1*}) \uparrow$	$\cos(z_2, \psi_{2*}) \uparrow$	$\cos(z_1, \psi_{1*}) \uparrow$	$\cos(z_2, \psi_{2*}) \uparrow$
SpecNet [14]	cp	0.99±0.00	0.99±0.00	0.98±0.00	0.98±0.00
	ps	0.96±0.01	0.94±0.03	0.88±0.05	0.88±0.05
	tp	0.98±0.02	0.97±0.02	0.94±0.04	0.72±0.10
	3Dc	0.89±0.11	0.79±0.17	0.85±0.11	0.77±0.13
	gs	0.90±0.10	0.90±0.10	0.91±0.10	0.91±0.10
	avg.	0.94±0.05	0.92±0.06	0.91±0.06	0.85±0.08
SpecNet2 [15]	cp	0.93±0.00	0.99±0.00	0.90±0.00	0.89±0.00
	ps	0.96±0.04	0.96±0.04	0.96±0.04	0.96±0.04
	tp	0.85±0.06	0.84±0.07	0.86±0.07	0.86±0.07
	3Dc	0.83±0.11	0.71±0.22	0.85±0.05	0.58±0.21
	gs	0.92±0.08	0.91±0.08	0.91±0.08	0.90±0.07
	avg.	0.90±0.06	0.88±0.08	0.90±0.05	0.84±0.08
Amo [50]	cp	0.98±0.00	0.96±0.00	0.98±0.01	0.99±0.01
	ps	0.95±0.02	0.93±0.04	0.91±0.03	0.91±0.04
	tp	0.97±0.03	0.95±0.04	0.93±0.06	0.81±0.06
	3Dc	0.80±0.08	0.85±0.09	0.84±0.09	0.79±0.07
	gs	0.92±0.08	0.91±0.08	0.90±0.08	0.89±0.07
	avg.	0.92±0.04	0.92±0.05	0.91±0.05	0.88±0.05
Ours	cp	1.00±0.00	1.00±0.00	0.99±0.00	0.99±0.00
	ps	0.99±0.02	0.99±0.02	0.93±0.07	0.83±0.07
	tp	0.98±0.02	0.96±0.04	0.98±0.01	0.91±0.05
	3Dc	0.96±0.04	0.96±0.04	0.97±0.02	0.96±0.02
	gs	0.94±0.07	0.93±0.07	0.94±0.04	0.93±0.06
	avg.	0.97±0.03	0.97±0.03	0.96±0.03	0.92±0.04

TABLE II
SPECTRAL BASIS APPROXIMATION ACCURACIES OF COMPARED METHODS REGARDING THE FIRST EIGHT AND 16 SPECTRAL BASES ON TWO EXPERIMENTAL SETTINGS, I.E., THE MODELS TRAINED ON FAUST AND TESTED ON FAUST (F-F) AND SCAPE (F-S).
† INDICATES MODEL TRAINING USING OUR EMBEDDING CRITERION [SEE (2)]. * INDICATES USING OUR NETWORK ARCHITECTURE

	F-F		F-S	
	ACD ₁₆ ↑	ACD ₈ ↑	ACD ₁₆ ↑	ACD ₈ ↑
SpecNet [14]	0.53±0.05	0.65±0.04	0.52±0.05	0.63±0.04
SpecNet2 [15]	0.46±0.04	0.57±0.03	0.44±0.04	0.55±0.03
Amo [50]	0.73±0.20	0.86±0.14	0.64±0.18	0.76±0.14
Amo* [50]	0.78±0.18	0.91±0.11	0.72±0.20	0.84±0.13
GCN [7]	0.51±0.03	0.67±0.04	0.51±0.03	0.68±0.03
SGC [9]	0.41±0.16	0.52±0.15	0.42±0.16	0.54±0.15
LightGCN [41]	0.41±0.17	0.54±0.16	0.43±0.17	0.56±0.16
GAE [32]	0.33±0.05	0.36±0.05	0.34±0.06	0.38±0.05
GAT [34]	0.36±0.02	0.45±0.03	0.36±0.02	0.46±0.03
GATv2 [33]	0.33±0.02	0.43±0.03	0.33±0.02	0.42±0.02
TokenGT [35]	0.29±0.04	0.46±0.05	0.30±0.04	0.46±0.05
RT [36]	0.38±0.03	0.52±0.04	0.38±0.02	0.50±0.03
GCN† [7]	0.59±0.04	0.75±0.05	0.55±0.05	0.72±0.07
SGC† [9]	0.54±0.06	0.74±0.05	0.50±0.05	0.70±0.06
LightGCN† [41]	0.36±0.04	0.48±0.05	0.42±0.03	0.52±0.04
GAT† [34]	0.47±0.05	0.71±0.05	0.44±0.05	0.68±0.06
GATv2† [33]	0.50±0.05	0.73±0.05	0.46±0.04	0.68±0.04
RT† [36]	0.48±0.04	0.66±0.05	0.47±0.04	0.65±0.06
Ours	0.81±0.07	0.94±0.06	0.77±0.11	0.89±0.09

SCAPE (F-S). Table II reports the ACD regarding the first eight and 16 spectral bases. The proposed approach achieves an ACD₈ ranging from 0.89 to 0.94. When considering the first 16 spectral bases, the ACD₁₆ is over 0.77 in both experimental settings. We notice that when applied to the LGC-based model,

TABLE III

SPECTRAL BASIS APPROXIMATION ACCURACIES OF COMPARED DEEP EIGENDECOMPOSITION METHODS REGARDING THE FIRST EIGHT AND 16 SPECTRAL BASES IN OBJECT-DEPENDENT AND INDEPENDENT SETTINGS ON THE TOSCA DATASET

	Dependent		Independent	
	ACD ₈ ↑	ACD ₁₆ ↑	ACD ₈ ↑	ACD ₁₆ ↑
SpecNet [14]	0.60±0.05	0.49±0.05	0.57±0.02	0.46±0.04
SpecNet2 [15]	0.50±0.04	0.44±0.03	0.48±0.03	0.41±0.02
Amo [50]	0.63±0.16	0.53±0.15	0.59±0.17	0.50±0.16
Ours	0.86±0.08	0.73±0.08	0.84±0.03	0.67±0.04

the proposed spectral embedding criterion exhibits higher variance in the F-S setting compared to its application to GCN, SGC, and other models. The relatively large variance can be attributed to the accumulation of approximation errors in spectral bases with higher Dirichlet energies. Although our method exhibits a greater variance in ACD₈ and ACD₁₆ in the F-S setting, our method still shows significant performance gains over the compared methods. On the other hand, the compared methods fail in approximating most of the spectral bases. As shown in Fig. 4, the similarities in the first eight spectral bases are above 0.80 in the F-F setting. The proposed model is feasible for recovering multiple spectral bases of graph matrices, avoiding being stuck to the dominant eigenvector.

Fig. 2(a) shows the sampled spectral bases Z by the proposed SBL. The proposed approach generates lined-up spectral bases without eigenvector switching or sign flipping. We report the discrete Dirichlet energy $\gamma_i = z_i^T L z_i$, $1 \leq i \leq k$. It is interesting to note that the generated bases are not sorted according to the magnitude of Dirichlet energies, while the energy distributions are similar on all sampled shapes [see Fig. 2(b)]. The generated spectral bases are consistent across graphs, indicating that the proposed SBL has the potential to select consistent spectral bases across graphs with identical intrinsic structures. Although there is no constraint for eigenvalue ordering in the loss function $\mathcal{L}$ [see (2)], the selected spectral bases are at similar frequency bands across graphs. The estimated spectral bases satisfy the orthogonality and fundamental properties of eigenvectors, as shown in Fig. 5, where both $Z^T Z$ and $Z^T L Z$ approach diagonal matrices. Note that there is no additional QR-based orthogonalization in the training or the postprocessing. The LWs in the proposed SBL are effective in producing orthogonal spectral bases of the graph Laplacian matrix.

Fig. 1(b) and (c) illustrates three pairs of predicted and ground-truth spectral bases $(z_m, \psi_{m'})$ of sampled meshes in curves and pseudocolors on 3-D meshes from the FAUST dataset. The predicted spectral bases are consistent with their counterparts in the conventional eigendecomposition. The similarity between the predicted and the conventional QR iteration-based method is over 0.97 on all reported spectral bases.

3) *TOSCA*: As to the TOSCA dataset, the proposed approach achieves spectral basis approximation accuracies of 0.73 and 0.86 regarding ACD₁₆ and ACD₈, respectively (see Table III). We conduct object-dependent experiments on three

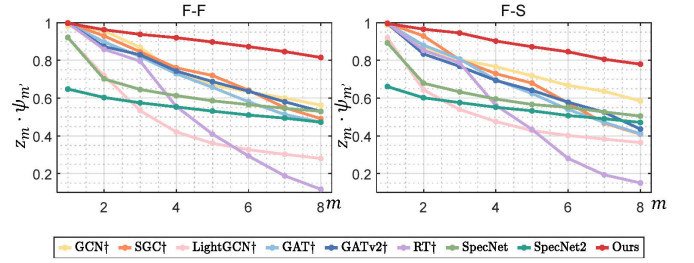


Fig. 4. Spectral basis similarity $z_m \cdot \psi_{m'} |_{1 \leq m \leq 8}$ between the predicted z_m by compared methods and the ground truth $\psi_{m'}$, including GCN [7], SGC [9], LightGCN [41], GAT [34], GATv2 [33], RT [36], SpecNet [14], SpecNet2 [15], and ours, regarding the first eight spectral bases on the FAUST and SCAPE datasets. † indicates using our embedding criterion [see (2)].

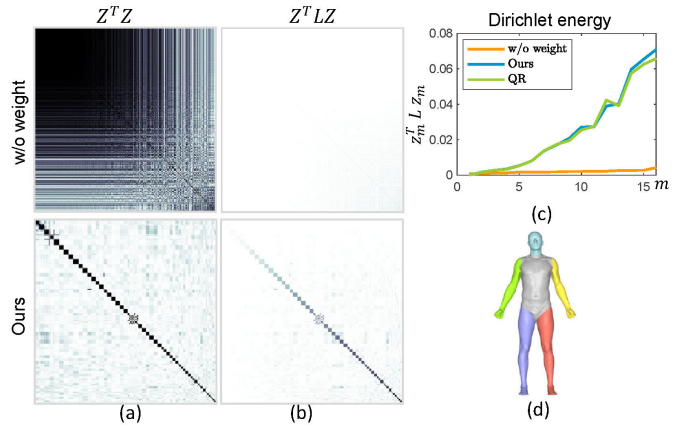


Fig. 5. (a) $Z^T Z$ with and without using the LW matrix. (b) $Z^T L Z$ with and without using the LW matrix. (c) Discrete Dirichlet energy $z_m^T L z_m$ regarding 16 approximated spectral bases. (d) Node labels used in the supervised learning of compared GCN variants.

types of meshes, i.e., cat, dog, and wolf, where the training and testing datasets have the same types of meshes. We also conduct object-independent experiments on the horse meshes, which are only used for testing. We report the spectral basis approximation accuracies regarding the first eight and 16 spectral bases (see Table III and Fig. 6). On both object-dependent and independent experiments, the proposed approach shows its capacity for spectral embedding with an ACD₈ of 0.86 and 0.84, respectively. Fig. 7 shows the spectral approximation accuracies on four types of meshes in the TOSCA datasets. The spectral approximation accuracies are over 0.81, 0.87, 0.88, and 0.84 regarding the cat, dog, wolf, and horse on the first eight spectral bases. Note that even when there are no horse meshes for training, the model learned from other types of animal meshes enables reasonable spectral basis approximation for horse meshes.

C. Comparison

1) *Graph Embedding Methods*: Table II reports quantitative comparisons regarding the spectral basis approximation accuracies by compared methods, including GCN [7], SGC [9], LightGCN [41], GAE [32], and self-attention neural networks [33], [34], [35], [36] regarding the first eight and 16 spectral bases on FAUST and SCAPE datasets. As to the supervised

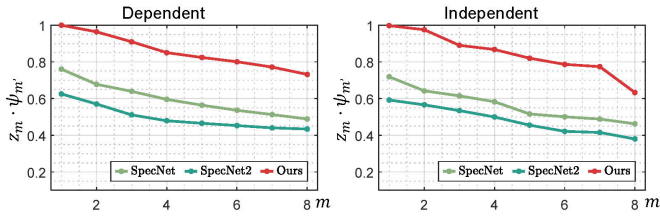


Fig. 6. Spectral basis approximation accuracies by the proposed SBL and compared methods, i.e., SpecNet [14], SpecNet2 [15], and ours, regarding the first eight spectral bases on the object-dependent and independent settings of the TOSCA dataset.

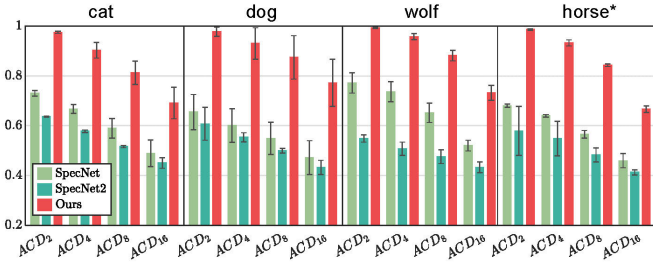


Fig. 7. Spectral basis approximation accuracies of compared deep eigendecomposition methods regarding the ACD on four types of meshes in TOSCA by compared SpecNet [14] and SpecNet2 [15]. * indicates the object-independent setting.

models, including GCN, SGC, LightGCN, GAT, and RT, we use the node labels regarding the shape segmentation map [see Fig. 5(d)] to train the classifier. We also train the compared models in an unsupervised manner by using the proposed embedding loss $\mathcal{L}$ [see (2)]. The proposed approach shows performance gains over compared methods in both supervised and unsupervised settings. For instance, the proposed SBL outperforms the compared methods by 0.30 over the GCN, 0.40 over the SGC, 0.40 over the LightGCN, and 0.48 over the GAE in the F-F setting regarding the ACD_{16} . As we can see, the supervised GCN variants learn the task-specific graph embedding, though irrelevant to the eigenstructures of the graph matrices.

The self-attention graph neural networks, including GAT [33], [34], TokenGT [35], and RT [36], learn task-specific graph embedding by exploiting global graph correlations and relational structures (see Table II). However, the graph embedding using the self-attention does not coincide with the eigendecomposition of the graph Laplacian. For instance, the ACD_8 ranges from 0.43 to 0.52 in the F-F setting, and 0.42 to 0.50 in the F-S setting, compared to self-attention methods. In contrast, the proposed SBL exploits linear algebra analogous to iterative power deflation, showing superiority over the compared self-attention-based graph embedding.

We compare with the graph embedding methods trained in an unsupervised manner by using the proposed embedding loss [see (2)]. The compared variants of GCN failed to generate spectral bases. We think the nonlinear activations in the shallow GCNs are limited to spectral basis approximation. By removing the classifier from the SGC and the LightGCN, there are no LWs, and the power iteration regarding the graph matrix is restricted to the dominant eigenvectors. The power

iteration [9] and polynomials [41] are limited to generating multiple spectral bases. The unsupervised GAE learned by affinity matrix reconstruction generated embeddings irrelevant to spectral bases. In contrast, the proposed approach exploits cascaded LGCs for multifrequency spectral basis approximation.

2) *Deep Eigendecomposition Methods*: We compare with deep eigendecomposition models, including SpecNet [14], SpecNet2 [15], and the amortized eigendecomposition (Amo) method [50]. Amo relies on eigenloss and QR factorization for eigenvector approximation. Since Amo introduces an eigenloss function for spectral decomposition without a dedicated network architecture, we conduct comparisons on two models using the eigenloss: 1) the three-layered GCN used in the Amo [50] and 2) the LGC network denoted as Amo*.

As to the toy data, the proposed SBL outperforms compared methods in all five types of toy data in the *single* and *mixed* experimental settings, as shown in Table I. For instance, the proposed SBL has performance gains of 0.03 and 0.05 over SpecNet regarding the first two nontrivial spectral bases in the *single* setting. We observe similar performance gains in the *mixed* setting. We conduct the comparison on the FAUST and SCAPE datasets, as shown in Table II. The proposed SBL outperforms the FCN-based eigendecomposition approaches by 0.26 [14] and 0.34 [15] regarding ACD_8 in the F-S setting. The baseline Amo using the GCN and eigenloss achieves ACD_{16} of 0.73 on the F-F setting and 0.64 on the F-S setting. Amo* leads to noticeable improvements, where ACD_{16} increases by 0.05 on the F-F setting and 0.08 on the F-S setting, respectively. Our approach produces better performances, underscoring the synergistic effect of the architectural design and the spectral embedding criterion. Note that Amo enforces the orthogonality via QR decomposition without specialized architecture design. In contrast, we present an LGC network to discover the underlying eigenstructures. The experimental results validate that our approach effectively approximates a set of orthogonal spectral bases while avoiding the computational cost involved in matrix factorization. As to TOSCA data, the proposed approach shows performance gains on all four types of meshes. Fig. 8 shows the projection of the first approximated nontrivial eigenvector by comparing deep eigendecomposition methods. As we can see, the proposed SBL has a small tail as to the projection vector $\{z \cdot \psi_i\}_{i=1}^{10}$, which indicates the orthogonality of the approximated spectral bases. In contrast to the FCN-based embedding [14], [15], we train and evaluate the eigendecomposition without additional QR iteration or affine transformation for orthogonalization in the training process. However, the proposed approach produces spectral bases in an end-to-end manner, outperforming the compared deep spectral embedding methods by taking advantage of the power iteration with deflation.

We provide comparisons with the deep learning models, including SpecNet [14] and SpecNet2 [15], regarding the computational complexity, as shown in Fig. 9. SpecNet uses two fully connected layers, one linear layer, and a QR decomposition for spectral bases approximation, where the computational time increases rapidly with the number of nodes. SpecNet2 approximates the spectral basis using two

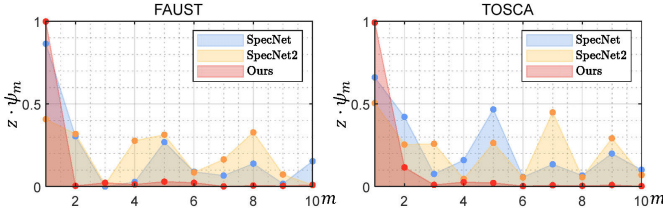


Fig. 8. Projection $\{z \cdot \psi_m^10\}$ of the first approximated nontrivial spectral basis with the ground truth bases by compared deep eigendecomposition methods, i.e., SpecNet [14], SpecNet2 [15], and ours.

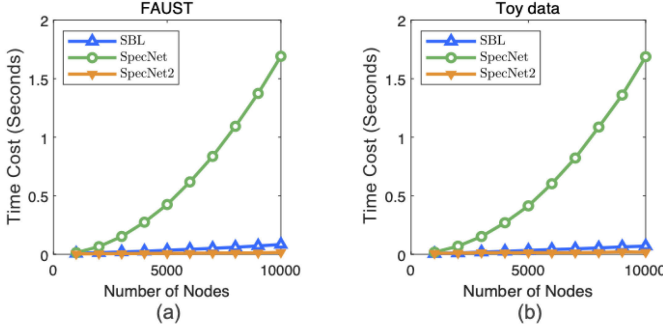


Fig. 9. Time cost of SpecNet [14], SpecNet2 [15], and proposed SBL on graphs of different sizes on (a) FAUST and (b) toy data.

TABLE IV

ABLATION STUDY REGARDING LGC-BASED SPECTRAL EMBEDDING, THE LW, AND THE SR ON FAUST DATASET

LGC	LW	SR	ACD ₈ ↑	ACD ₁₆ ↑
✓			0.47±0.04	0.29±0.02
✓			0.73±0.01	0.53±0.03
✓	✓		0.91±0.09	0.77±0.12
✓	✓	✓	0.94±0.06	0.81±0.07

fully connected layers and one linear layer, incurring a lower computational cost than SpecNet and the proposed SBL. The proposed SBL uses three fully connected layers and five LGC layers for spectral bases approximation. Owing to the use of more fully connected layers and sparse multiplication operations than SpecNet2, the SBL method entails a slightly higher computational cost than SpecNet2, yet it remains lower than that of SpecNet.

D. Ablation Study

We conduct ablation studies to verify the effectiveness of the LGC-based spectral embedding, the LWs, and the SR.

1) *Effectiveness of LGC-Based Embedding*: The first variant model in Table IV uses the FCN-based descriptor learning, which is learned by minimizing $\mathcal{L}(\tilde{F})$. We add LGC-based embedding modules without LW matrices, which are learned by minimizing $\mathcal{L}(\tilde{Z})$. The cascaded LGCs greatly improve the spectral basis accuracy. As shown in Table IV, the ACD₈ increases by 0.26 and ACD₁₆ increases by 0.24.

2) *Effectiveness of LW*: Table IV shows the spectral basis approximation performance gains when using LWs. The LW matrix is optimized to simulate the power method with deflation for spectral basis approximation with the ACD₁₆ of

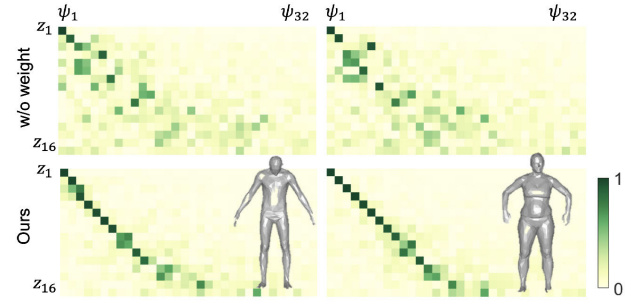


Fig. 10. Similarity matrix between the estimated spectral bases Z and Ψ with and without using the LW matrix on two testing shapes from the FAUST dataset. The meshes are plotted at the right corner.

TABLE V

ABLATION STUDY OF FUNDAMENTAL PROPERTIES OF EIGENVECTORS AND ORTHOGONALITY CONSTRAINTS ON MODELS TRAINED ON FAUST AND TESTED ON FAUST (F-F) AND SCAPE (F-S)

$\mathcal{L}_{ev}$	$\mathcal{L}_{dia}$	$\mathcal{L}_{ort}$	F-F		F-S	
			ACD ₈ ↑	ACD ₁₆ ↑	ACD ₈ ↑	ACD ₁₆ ↑
✓	✓		0.51±0.28	0.28±0.30	0.50±0.33	0.31±0.24
✓		✓	0.74±0.13	0.62±0.17	0.70±0.08	0.60±0.14
	✓	✓	0.91±0.10	0.76±0.19	0.87±0.13	0.72±0.22
✓	✓	✓	0.94±0.06	0.81±0.08	0.89±0.09	0.77±0.11

0.77. The approximated spectral bases satisfy the fundamental property of eigenvectors. On the contrary, when removing the LW matrix from the spectral embedding module, the model cannot generate the orthogonal spectral bases [see Fig. 5(a)]. However, the power iteration of the graph matrix without LWs is limited to discovering the spectral basis except the dominant one, as shown in Fig. 10. The LW matrices show capacities of deflation by removing correlations with the dominant eigenvectors.

3) *Effectiveness of Refiner*: The proposed SBL introduces the SR and avoids additional sorting in the postprocessing. The score sorting from the full spectral embedding is capable of finding multiple spectral bases. However, the straightforward score sorting cannot remove the correlated eigenvectors. The proposed SBL with the SR forces orthogonality compared with variant models. The estimated spectral bases satisfy the fundamental properties of eigenvectors with an ACD₁₆ of 0.81, outperforming all compared variants. Note that the proposed SR enables spectral basis estimation without additional transformation or orthogonalization in the postprocessing.

4) *Effectiveness of Fundamental Properties of Eigenvectors and Orthogonality Constraints*: We provide an ablation study to show the individual contributions of fundamental properties of eigenvectors and orthogonality constraints in the spectral basis approximation loss [see (2)] to the overall performance (see Table V). When the orthogonality constraint $\mathcal{L}_{ort}$ is absent, the model achieves ACD₈ of 0.51 on the F-F setting and 0.50 on the F-S setting, while ACD₁₆ drops significantly to 0.28 and 0.31, respectively. This indicates that without orthogonal regularization, the learned spectral bases only approximate a limited number of eigenvectors and even

TABLE VI
SPECTRAL BASIS APPROXIMATION ACCURACIES WITH DIFFERENT
VALUES OF α_2 ON THE FAUST DATASET

	0.1	0.2	0.5	1
$ACD_8 \uparrow$	0.89 ± 0.06	0.94 ± 0.06	0.91 ± 0.06	0.84 ± 0.05
$ACD_{16} \uparrow$	0.68 ± 0.07	0.81 ± 0.07	0.78 ± 0.08	0.69 ± 0.06

degenerate to dominant eigenvectors. When removing $\mathcal{L}_{dia}$, ACD_8 decreases by 0.20 on the F-F setting and 0.19 on the F-S setting, demonstrating that the diagonal similarity transformation constraint enhances the spectral bases approximation performances. When combining $\mathcal{L}_{ev}$, we notice that spectral approximation accuracy of ACD_8 and ACD_{16} increases on both F-F and F-S settings. However, the constraints of fundamental properties of eigenvectors, diagonal similarity transformation, and orthogonality are complementary, and the combination of these constraints produces the best performance.

E. Parameter Analysis

We use hyperparameters α_1 and α_2 to balance the orthogonal regularization and the fundamental properties of eigenvectors [see (2)]. We set α_1 to 1 for the same order of magnitude of fundamental eigendecomposition losses of $\mathcal{L}_{ev}$ and $\mathcal{L}_{dia}$. As shown in Table VI, ACD reaches a local maximum when α_2 is set to 0.2. In our experiments, the graph Laplacian matrices are of similar scales with approximately 1k nodes. The data terms of $\mathcal{L}_{ev}$ and $\mathcal{L}_{dia}$, as well as the orthogonal regularization term of $\mathcal{L}_{ort}$, have similar magnitudes across different datasets. Thus, we employed the same set of hyperparameters on all datasets.

To analyze the effectiveness of LGC depth on the spectral basis approximation, we compare our model trained using different numbers of LGCs, as shown in Fig. 11. The ACD_{16} increases from 0.69 to 0.81 when the depth grows from 1 to 5 regarding the first eight spectral bases on the FAUST dataset. The approximated spectral bases produced by the shallow model have apparent correlations with dominant eigenvectors, where nonzero entries occur below the diagonal. In contrast, the deeper linear GCN shows performance gains by removing correlations with dominant eigenvectors for discovering multiple spectral bases. In our experiments, we observe no significant performance degradation regarding our parameter settings.

F. Convergence Analysis

We provide analysis of convergence and learning behavior, as shown in Fig. 12. We observe that the loss function on both the training and validation datasets decreases rapidly during the first two epochs. Then the decrease rate slows down. Following approximately 20 epochs of gradual decline, the downward trend further slows down, eventually stabilizing after approximately 30 epochs. In our experiments, we train the proposed SBL for 35 epochs. We also report the learning behavior regarding individual loss terms, including $\mathcal{L}_{ev}$, $\mathcal{L}_{dia}$, and $\mathcal{L}_{ort}$. All three conditions tend to converge in learning the

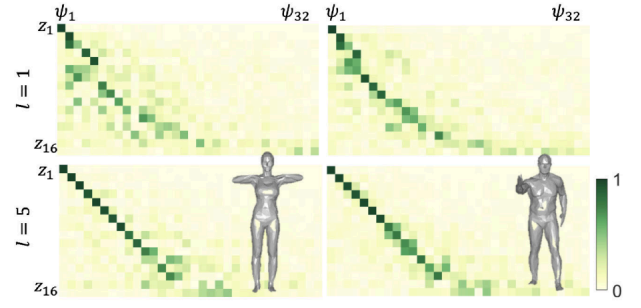


Fig. 11. Similarity matrix of the estimated spectral bases Z with the ground truth Ψ using the cascaded LGC with depths $l = 1$ and $l = 5$ on two testing shapes from the FAUST dataset. The meshes are plotted at the right corner.

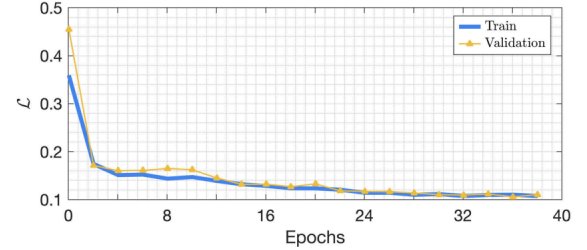


Fig. 12. Convergence curves on faust dataset.

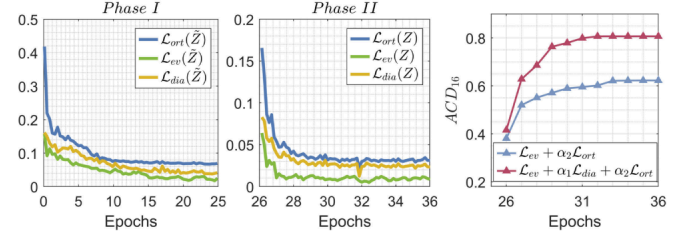


Fig. 13. (a) Left and middle are curves of the individual loss terms, including $\mathcal{L}_{ort}$, $\mathcal{L}_{ev}$, and $\mathcal{L}_{dia}$, during the training process. Right is ACD_{16} of the proposed model with and without $\mathcal{L}_{dia}$ on FAUST during the training process.

SBL, as shown in Fig. 13(a). We present the spectral basis approximation accuracy in the second training phase when using only the orthogonality and eigenvector directionality constraints [see Fig. 13(b)]. We observe that the model without the $\mathcal{L}_{dia}$ converged at a lower ACD_{16} value of 0.62 in the 33rd epoch, and exhibited slightly slower convergence compared to the model using $\mathcal{L}_{dia}$. We think that $\mathcal{L}_{dia}$ enhances the model by balancing the orthogonality and fundamental properties of eigenvectors.

G. Graph Matching

The SBL is feasible for producing consistent spectral bases across graphs for correspondence computation in a low-dimensional embedding space, relieving additional feature projection in the FM-based methods [5], [51]. The affine transformation $P = \arg \min_P \|Z_r P - Z_t\|_2^2$. Z_r and Z_t denote the approximated spectral bases of the reference graph $\mathcal{G}_r$ and the target graph $\mathcal{G}_t$, respectively. The pointwise map $Q \in \mathbb{R}^{n \times n}$ is computed as: $Q = Z_r P(Z_t)^T$. The correspondence using the spectral bases predicted by the proposed SBL achieves

TABLE VII

CORRESPONDENCE ACCURACIES AGD ON FAUST AND SCAPE. † INDICATES USING THE PROPOSED EMBEDDING LOSS [SEE (2)]. ‡ DENOTES USING APPROXIMATED SPECTRAL BASES Z , WHILE § REPRESENTS THE USE OF Ψ OBTAINED VIA QR ITERATION

	GCN [7]	GCN† [7]	SGC [9]	SGC† [9]	FM‡ [5]	FMNet‡ [51]	FM§ [5]	FMNet§ [51]	Ours
F-F	0.20±0.11	0.19±0.08	0.21±0.1	0.20±0.11	0.21±0.14	0.20±0.13	0.23±0.12	0.20±0.07	0.15±0.13
F-S	0.34±0.11	0.32±0.09	0.34±0.11	0.34±0.12	0.36±0.12	0.37±0.15	0.34±0.11	0.32±0.09	0.26±0.08

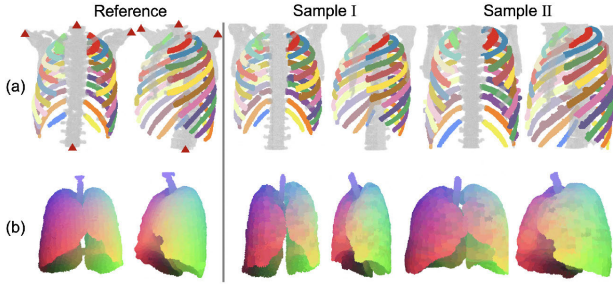


Fig. 14. Dense correspondence of (a) rib cages and (b) lungs in two views. Pseudocolors indicate corresponding supervoxels. Four boundary supervoxels are plotted in red triangles.

the AGD of 0.15 and 0.26 in F-F and F-S experimental settings, outperforming the conventional eigendecomposition and FM-based methods (see Table VII). The embedding by the GCN [7] and SGC [9] tailored for the classification task is limited to establishing reliable dense correspondence. As to the unsupervised SGC† using the proposed spectral embedding criteria, the lack of LW matrices in the embedding modules makes it difficult to generate spectral bases for shape matching. On the other hand, the LW matrices in the proposed SBL contribute to power deflation to approximate spectral bases, achieving performance gains for dense correspondence.

H. Anatomical Structure Correspondence

The proposed SBL is also feasible to establish anatomical structure correspondence on volumetric chest CTs [20]. The proposed SBL produces the spectral bases Z of supervoxel graphs of lungs and bony structures. As shown in Fig. 14, the proposed SBL is feasible to capture the correspondence of lungs and ribs when confronted with intersubject structure variations and different respiratory stages. However, the spectral embedding from the SBL captures the rib arrangement and facilitates the reliable correspondence of individual ribs.

We provide quantitative results of rib segmentation (see Table VIII). We compare with traditional B-spline registration [52]-based segmentation, GCN [7], SGC [9], and FM [5] methods. Our method outperforms the B-spline registration method on most of the ribs regarding the DSC. We note that the B-spline registration approach tends to be trapped in a local extremum due to the difficulty in distinguishing ribs. In contrast, our SBL framework has a comprehensive understanding of the underlying anatomical topology, thereby reducing confusion between neighboring ribs. Our method also outperforms the GCN-based methods [7], [9], and the FM

TABLE VIII

SEGMENTATION RESULTS OF THE UPPER (1ST–3RD), MIDDLE (4TH–9TH), AND LOWER (10TH–12TH) RIBS REGARDING THE DSC ON THE RIBFRAC DATASET. † INDICATES USING THE PROPOSED EMBEDDING LOSS [SEE (2)]. ‡ DENOTES USING APPROXIMATED SPECTRAL BASES Z

	GCN† [7]	SGC† [9]	FM‡ [5]	B-spline [52]	Ours
Upper	0.62±0.13	0.46±0.12	0.48±0.17	0.65±0.08	0.76±0.10
Middle	0.52±0.07	0.37±0.07	0.40±0.09	0.44±0.13	0.57±0.07
Lower	0.59±0.05	0.47±0.03	0.49±0.11	0.42±0.08	0.69±0.04

TABLE IX

SKELETON-BASED ACTION RECOGNITION REGARDING THE RECALL ON THE DRIVE&ACT DATASET [44]

	Recall(%)
ST-GCN [53]	45.34
ST-GCN+KRM [49]	47.63
Ours	47.66

method [5] in generating dense correspondence and segmentation label transfer.

I. Action Recognition

We validate our method on the Drive&Act dataset [44] for skeleton-based action recognition (see Table IX). We use the same experimental protocol and the skeletal data of 34 action categories as [49]. We benchmark our results against the ST-GCN [53] and ST-GCN+KRM [49]. In contrast to ST-GCN's graph construction paradigm, which treats each frame's skeletal joints as an individual graph, we propose a unified graph representation that integrates all skeletal frames within a temporal segment into a single graph. To balance global and local information, our framework uses the spectral bases produced by the SBL from the human action graph, together with the original node input features for classification. The classifier comprises two fully connected layers followed by a global pooling layer. We report an average class recall of 47.66%, with an improvement of 2.32% over the ST-GCN [53], and comparable to the ST-GCN enhanced with KRM [49]. The proposed approach enhances the model's capacity to capture holistic spatiotemporal dependencies and structural dynamics of human actions.

V. CONCLUSION

This article has presented a novel unsupervised SBL framework for generalized eigendecomposition of the graph Laplacian and spectral embedding. The proposed approach benefits from cascaded LGCs in an analog of the iterative

power deflation to estimate multifrequency spectral bases with a theoretical bound. Our approach facilitates consistent spectral basis discovery across graphs, enabling graph matching without additional eigenvector switching or sign flipping. Extensive experimental results showcase the effectiveness of the proposed SBL in generalized eigendecomposition and spectral embedding compared with existing deep graph embedding and eigendecomposition methods.

In the current work, the computational complexity depends on the bandwidth of the graph matrix. The computational complexity will be enlarged when confronted with a dense affinity matrix. The computationally-efficient extension regarding dense graph matrices still deserves further study. However, we observe drops in spectral approximation accuracies with increasing Dirichlet energies. Although the proposed SBL has shown superior performance over the compared deep graph embedding and eigendecomposition methods, further investigations on spectral embedding learning are deserved to handle error accumulations with increasing Dirichlet energies. Our approach achieves state-of-the-art eigenvector approximation accuracy, owing to its specialized network architecture and the spectral embedding criterion. Unlike adapting general-purpose foundation models to eigendecomposition problems, our approach excels at capturing the underlying eigenstructures of graph matrices. In future work, we will apply the proposed SBL to the development of graph foundation models in a variety of downstream tasks. However, our current method relies on handcrafted graph node features. We note the emergence of foundation models capable of extracting more expressive representations from raw data. In future work, we will explore integrating the features produced by the pretrained foundation models to enhance the approximation of spectral bases.

REFERENCES

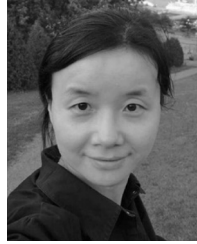
- [1] M. Belkin and P. Niyogi, "Laplacian eigenmaps for dimensionality reduction and data representation," *Neural Comput.*, vol. 15, no. 6, pp. 1373–1396, Jun. 2003.
- [2] A. Y. Ng, M. I. Jordan, and Y. Weiss, "On spectral clustering: Analysis and an algorithm," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 14, 2001, pp. 849–856.
- [3] F. De Goes, S. Goldenstein, and L. Velho, "A hierarchical segmentation of articulated bodies," *Comput. Graph. Forum*, vol. 27, no. 5, pp. 1349–1356, Jul. 2008.
- [4] V. Jain, H. Zhang, and O. M. van Kaick, "Non-rigid spectral correspondence of triangle meshes," *Int. J. Shape Model.*, vol. 13, no. 1, pp. 101–124, 2007.
- [5] M. Ovsjanikov, M. Ben-Chen, J. Solomon, A. Butscher, and L. Guibas, "Functional maps: A flexible representation of maps between shapes," *ACM Trans. Graph.*, vol. 31, no. 4, p. 30, 2012.
- [6] A. Nasikun, C. Brandt, and K. Hildebrandt, "Fast approximation of Laplace-Beltrami eigenproblems," *Comput. Graph. Forum*, vol. 37, no. 5, pp. 121–134, Aug. 2018.
- [7] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," in *Proc. Int. Conf. Learn. Represent.*, 2017, pp. 1–8.
- [8] M. Defferrard, X. Bresson, and P. Vandergheynst, "Convolutional neural networks on graphs with fast localized spectral filtering," in *Proc. Adv. Neural Inf. Process. Syst.*, 2016, pp. 3844–3852.
- [9] F. Wu, T. Zhang, A. H. Souza, C. Fifty, T. Yu, and K. Q. Weinberger, "Simplifying graph convolutional networks," in *Proc. Int. Conf. Mach. Learn.*, 2019, pp. 6861–6871.
- [10] C. E. Priebe, C. Shen, N. Huang, and T. Chen, "A simple spectral failure mode for graph convolutional networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 11, pp. 8689–8693, Nov. 2022.
- [11] R. Marin, M. Rakotosaona, S. Melzi, and M. Ovsjanikov, "Correspondence learning via linearly-invariant embedding," in *Proc. Adv. Neural Inf. Process. Syst.*, 2020, pp. 1608–1619.
- [12] Y. Bengio, J.-F. Paiement, P. Vincent, O. Delalleau, N. L. Roux, and M. C. Ouimet, "Out-of-sample extensions for LLE, isomap, MDS, eigenmaps, and spectral clustering," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 16, 2003, pp. 177–184.
- [13] I. Ben-Shaul, L. Bar, and N. Sochen, "Solving the functional eigenproblem using neural networks," 2020, *arXiv:2007.10205*.
- [14] U. Shaham, K. Stanton, H. Li, B. Nadler, R. Basri, and Y. Kluger, "SpectralNet: Spectral clustering using deep neural networks," in *Proc. Int. Conf. Learn. Represent.*, 2018.
- [15] Z. Chen, Y. Li, and X. Cheng, "SpecNet2: Orthogonalization-free spectral embedding by neural networks," *Math. Sci. Mach. Learn.*, vol. 190, pp. 33–48, 2022.
- [16] O. Streicher, I. Cohen, and G. Gilboa, "BASIS: Batch aligned spectral embedding space," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 10396–10405.
- [17] F. Bogo, J. Romero, M. Loper, and M. J. Black, "FAUST: Dataset and evaluation for 3D mesh registration," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2014, pp. 3794–3801.
- [18] D. Anguelov, P. Srinivasan, D. Koller, S. Thrun, J. Rodgers, and J. Davis, "SCAPE: Shape completion and animation of people," *ACM Trans. Graph.*, vol. 24, pp. 408–416, Jan. 2005.
- [19] A. M. Bronstein, M. M. Bronstein, and R. Kimmel, *Numerical Geometry of Non-Rigid Shapes* (Monographs in Computer Science). New York, NY, USA: Springer, 2009.
- [20] L. Jin et al., "Deep-learning-assisted detection and segmentation of rib fractures from CT scans: Development and validation of FracNet," *eBioMedicine*, vol. 62, Dec. 2020, Art. no. 103106.
- [21] (2023). *MATLAB-Toolbox-for-Dimensionality-Reduction*. [Online]. Available: <https://github.com/UMD-ISL/Matlab-Toolbox-for-Dimensionality-Reduction>
- [22] K. Pham and G. Chen, "Large-scale spectral clustering using diffusion coordinates on landmark-based bipartite graphs," in *Proc. 12th Workshop Graph-Based Methods Natural Lang. Process. (TextGraphs-12)*, 2018, pp. 28–37.
- [23] C. K. I. Williams and M. W. Seeger, "Using the Nyström method to speed up kernel machines," in *Proc. Adv. Neural Inf. Process. Syst.*, 2000, pp. 1–11.
- [24] M.-A. Belabbas and P. J. Wolfe, "On landmark selection and sampling in high-dimensional data analysis," *Phil. Trans. Roy. Soc. A, Math., Phys. Eng. Sci.*, vol. 367, no. 1906, pp. 4295–4312, Nov. 2009.
- [25] J. Platt, "FastMap, MetricMap, and landmark MDS are all Nyström algorithms," in *Proc. Int. Conf. Artif. Intell. Statist.*, 2005, pp. 1–8.
- [26] L. Besson, "Finding structure with randomness probabilistic algorithms for approximate matrix decompositions research project report sparsity and compressed sensing course," EPFL, Lausanne, Switzerland, Tech. Rep., Feb. 2016.
- [27] K.-J. Bathe, "The subspace iteration method—Revisited," *Comput. Struct.*, vol. 126, pp. 177–183, Sep. 2013.
- [28] D. Yan, L. Huang, and M. I. Jordan, "Fast approximate spectral clustering," in *Proc. Knowl. Discovery Data Mining*, 2009, pp. 907–916.
- [29] X. Shen and F.-L. Chung, "Deep network embedding for graph representation learning in signed networks," *IEEE Trans. Cybern.*, vol. 50, no. 4, pp. 1556–1568, Apr. 2020.
- [30] S. Cao, W. Lu, and Q. Xu, "Deep neural networks for learning graph representations," in *Proc. AAAI*, vol. 30, 2016, pp. 1145–1152.
- [31] T. N. Kipf and M. Welling, "Variational graph auto-encoders," 2016, *arXiv:1611.07308*.
- [32] S. Pan, R. Hu, G. Long, J. Jiang, L. Yao, and C. Zhang, "Adversarially regularized graph autoencoder for graph embedding," in *Proc. 27th Int. Joint Conf. Artif. Intell.*, Jul. 2018, pp. 2609–2615.
- [33] S. Brody, U. Alon, and E. Yahav, "How attentive are graph attention networks?," in *Proc. Int. Conf. Learn. Represent.*, 2021.
- [34] P. Velickovic, G. Cucurull, A. Casanova, A. Romero, P. Liò, and Y. Bengio, "Graph attention networks," in *Proc. Int. Conf. Learn. Represent.*, 2018, pp. 1–6.
- [35] J. Kim et al., "Pure transformers are powerful graph learners," in *Proc. Adv. Neural Inf. Process. Syst.*, 2022, pp. 14582–14595.
- [36] C. Diao and R. Loynd, "Relational attention: Generalizing transformers for graph-structured tasks," 2022, *arXiv:2210.05062*.
- [37] C. Wang, S. Pan, G. Long, X. Zhu, and J. Jiang, "MGAE: Marginalized graph autoencoder for graph clustering," in *Proc. ACM CIKM*, 2017, pp. 889–898.
- [38] J. Park, M. Lee, H. J. Chang, K. Lee, and J. Y. Choi, "Symmetric graph convolutional autoencoder for unsupervised graph representation learning," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 6518–6527.

- [39] P.-E. Lu and C.-S. Chang, "Explainable, stable, and scalable graph convolutional networks for learning graph representation," 2020, *arXiv:2009.10367*.
- [40] G. Mishne, U. Shaham, A. Cloninger, and I. Cohen, "Diffusion nets," *Appl. Comput. Harmon. Anal.*, vol. 47, no. 2, pp. 259–285, Sep. 2019.
- [41] X. He, K. Deng, X. Wang, Y. Li, Y. Zhang, and M. Wang, "LightGCN: Simplifying and powering graph convolution network for recommendation," in *Proc. 43rd Int. ACM SIGIR Conf. Res. Develop. Inf. Retr.*, Jul. 2020, pp. 639–648.
- [42] B. Lévy and H. Zhang, "Spectral mesh processing," in *Proc. Int. Conf. Comput. Graph. Interact. Techn.*, 2009, pp. 108:1–108:12.
- [43] J. D. F. Annette, M. Burden, and R. L. Burden, *Numerical Analysis*. Boston, MA, USA: Cengage Learning, 2015.
- [44] M. Martin et al., "Drive&act: A multi-modal dataset for fine-grained driver behavior recognition in autonomous vehicles," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 2801–2810.
- [45] A. Sharma and M. Ovsjanikov, "Weakly supervised deep functional map for shape matching," 2020, *arXiv:2009.13339*.
- [46] M. Garland and P. S. Heckbert, "Simplifying surfaces with color and texture using quadric error metrics," in *Proc. Visualizat.*, Mar. 1998, pp. 263–269.
- [47] F. Tombari, S. Salti, and L. D. Stefano, "Unique signatures of histograms for local surface description," in *Proc. Eur. Conf. Comput. Vis.*, 2010, pp. 356–369.
- [48] R. Achanta, A. Shaji, K. Smith, A. Lucchi, P. Fua, and S. Süsstrunk, "SLIC superpixels compared to state-of-the-art superpixel methods," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 34, no. 11, pp. 2274–2282, Nov. 2012.
- [49] M. Tan et al., "Bidirectional posture-appearance interaction network for driver behavior recognition," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 8, pp. 13242–13254, Aug. 2022.
- [50] T. Li, M. Lin, Z. Shi, and J. Zhao, "Amortized eigendecomposition for neural networks," in *Proc. Adv. Neural Inf. Process. Syst.*, 2024, pp. 138809–138835.
- [51] O. Litany, T. Remez, E. Rodolà, A. Bronstein, and M. Bronstein, "Deep functional maps: Structured prediction for dense shape correspondence," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 5660–5668.
- [52] D. Rueckert, L. I. Sonoda, C. Hayes, D. L. G. Hill, M. O. Leach, and D. J. Hawkes, "Nonrigid registration using free-form deformations: Application to breast MR images," *IEEE Trans. Med. Imag.*, vol. 18, no. 8, pp. 712–721, Aug. 1999.
- [53] B. Yu, H. Yin, and Z. Zhu, "Spatio-temporal graph convolutional networks: A deep learning framework for traffic forecasting," in *Proc. 27th Int. Joint Conf. Artif. Intell.*, Jul. 2018, pp. 3634–3640.



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